Abdurrahmaan Baghdadi

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Portfolio: 25 projects, 5 live demos at **abdurrahmaan-baghdadi.github.io** | github.com/Abdurrahmaan-Baghdadi

Summary

M.S. Artificial Intelligence student at UT Austin researching interpretable pathology AI and building large language model (LLM) systems; 5+ years of research experience in machine learning, IoT security, and full-stack development.

Education & Certifications

Master of Science (MS): *Artificial Intelligence* | **University of Texas at Austin** Expected May 2027
Bachelor of Science (BS): *Computer Science, Minor in Cybersecurity* | **University of Texas at San Antonio** May 2024
Summa cum laude | GPA 3.93 | Dean's List | Honors Program | Distinguished Presidential Scholarship
Certifications: CompTIA Security+ (Jul 2024) | DeepLearning.AI: Supervised ML, Advanced Learning Algorithms, Math for ML
Relevant Coursework: Advances in Deep Learning, Deep Learning, Machine Learning, AI in Healthcare (MIMIC clinical data)

Research Experience

Research Assistant | *Pediatric Brain Tumor AI Lab, University of Texas at Austin (Prof. Ying Ding)* 03/2026 – Present

- Benchmark vision-language models (**WSI-LLaVA**, **Quilt-LLaVA**, **Concept Bottleneck Models**, **WSI-VQA**) for multimodal pathology to support pediatric brain tumor diagnosis from whole-slide images (WSIs)
- Run PICASSO's TopK **sparse autoencoder** on **TACC** supercomputers to compare **UNI2** and **Virchow2** foundation-model embeddings across 1,700 TCGA GBM/LGG slides, with patient-level splits, ablations, and **SLURM** jobs
- Preprocessed whole-slide images, co-authored a TACC allocation request to run **I2MoE** (multimodal mixture-of-experts), and designed publication figures for lab papers

Research Assistant, IoT Security | *IoT Lab and SCADA Lab, University of Texas at San Antonio* 06/2019 – 01/2025

- Cut equipment loss **25%** with a full-stack inventory system for up to 100 IoT devices (**Python**, **Flask**, **React**, **TypeScript**, **SQL**)
- Recovered forensic artifacts from IoT devices and automated firmware **NIST** compliance checks (**Autopsy**, **Emba**); captured IoT threat telemetry with honeypots (**Elastic Stack**, **ROCK NSM**); secured **9 Linux** VMs and SCADA lab infrastructure

Professional Experience

Student Technology Assistant | *LAITS, College of Liberal Arts, University of Texas at Austin* 09/2026 – Present

- Designed a four-stage helpdesk router (**similarity filter**, **embedding search**, **LLM reader**, **decision step**) that routes web-help questions to the correct page among 64, asks a clarifying question, or opens a support ticket
- Fine-tuned **arctic-embed-m-v1.5** with contrastive **MultipleNegativesRankingLoss** and ensembled it with **mxbai-embed-large-v1** for retrieval; added a self-hosted **Qwen 3.8 27B** reader on **Ollama**, directed with **Claude Code**
- Benchmarked 11 embedding models, rerankers, and LLM readers in pre-registered tests; raised users helped from **43% to 75%** and cut wrong pages from **40% to 6%** vs. the legacy system; deployed via **FastAPI** to **Kubernetes** (test)
- Built a full local **WordPress** stack (**Bedrock**, **Docker**, **Composer**) for containerizing college sites; partnered with the senior developer on **Woodpecker CI** and **Rancher** deployment; rebuilt faculty research-lab sites on the Labs multisite

Backend Developer | *JIAGIA Studios* 09/2026 – Present

- Co-develop a **REST API** (**Fastify**, **PostgreSQL**, **TypeScript**, **Zod**) with the lead developer for a research and art platform

Code Sensei & Camp Director | *Code Ninjas, Round Rock, TX* 01/2025 – 08/2025

- Grew program sales **13%** via operations and outreach; taught up to **15 students** ages 5 to 14 (ML, robotics, JavaScript, C#)

Selected Projects

- **Energy AI Hackathon, Mission Impossible (Jan 2026):** As sole programmer on an interdisciplinary team, built a stacked ensemble (**HistGradientBoosting**, **GradientBoosting**, **ExtraTrees**, **Ridge**) with **GroupKFold** cross-validation and physics-based features; reached an out-of-fold **R-squared of 0.9849** among 40 teams (sponsored by Phillips 66, Chevron, and ExxonMobil)
- **Sleep Staging from Apple Watch Data:** Built a **LightGBM** model with hidden Markov model (**HMM**) smoothing on 247 nights of heart-rate and motion data; improved Cohen's kappa in every cross-validation fold (**0.42 to 0.45**)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Swift, Java, PHP, C, C#, Bash, R, HTML/CSS
Machine Learning & AI: PyTorch, Hugging Face Transformers, LLMs, Retrieval-Augmented Generation (RAG), LoRA/QLoRA, Computer Vision, NLP, vLLM, Ollama, LightGBM, XGBoost, Scikit-learn, Pandas
Web, Mobile & Data: FastAPI, Fastify, Flask, React, Node.js, WordPress, SwiftUI, PostgreSQL, MySQL, MongoDB
Infrastructure & Security: Docker, Kubernetes, Rancher, SLURM, HPC (TACC), Git, GitHub Actions, AWS EC2, Linux, Wireshark, Elastic Stack, Autopsy